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Ferroelectric mechanism of croconic acid: A first-principles and Monte Carlo study

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The ferroelectric mechanism of croconic acid in terms of the electronic structure and the molecular structure was studied by first principles using the density functional theory with the generalized gradient approximation. The spontaneous polarization (P_s) was simulated by the Berry phase method. It is found that the large polarization originates from charge transfer due to the strong “push-pull” effect of electron-releasing and -withdrawing groups along the hydrogen bond. According to the characteristics of polarization of croconic acid, we constructed a one-dimensional ferroelectric Hamiltonian model to describe the ferroelectric properties of croconic acid. Based on the Hamiltonian model, the thermal properties of the ferroelectricity of croconic acid were studied by Monte Carlo method. The simulated Curie temperature is 756 K, and the spontaneous polarization keeps well temperature range stability up to 400 K. These results are in good agreement with the experimental data. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4813500>]

I. INTRODUCTION

Ferroelectric materials, which show spontaneous polarization and external electric field dependent polarization switching behaviors, have attracted intense interest due to their potential applications in piezoelectric transducer, ferroelectric tunnel junction (FTJ), capacitors, nonlinear optical devices, and memory devices.^{1–4} Although perovskite oxide ceramics are most widely used for those applications, organic ferroelectrics could be attractive alternatives due to their excellent properties, such as nontoxic, mechanical flexibility, and environmentally friendly processing as they are lead-free. Moreover, these mentioned properties turn to make it easy for single-component low-molecular-mass organic ferroelectric to fabricate flexible, lightweight, large-area, low-cost organic device by varied methods, such as spin-coating, spray-on, inkjet printing, and vapour-phase deposition techniques.¹ Up today, only a few single-component organic compounds are reported to exhibit ferroelectric properties.^{1,5} However, except thiourea, most of the reported single-component organic compounds possess a rather low dielectric constant. These compounds are not categorized into intrinsic ferroelectrics due to their ferroelectricity coming from the side-effect of a ferroelastic or structural instability rather than the dipole-dipole interactions.¹ It is still a great challenge to find new single-component organic compounds with intrinsic ferroelectric properties.

The recent research founded that the proton transfer plays an important role in the ferroelectric property of organic compounds. In 2010, the first single-component organic ferroelectrics of proton transfer type, croconic acid (4,5-dihydroxy-4-cyclopentene-1,2,3-trione, $\text{H}_2\text{C}_5\text{O}_5$) was

reported by Horiuchi *et al.*,⁶ which brings a breakthrough in the exploration of new organic ferroelectrics. Since the collective site-to-site transfer of protons dramatically reduces the energy needed in switching of polarization, the proton-transfer-type materials have small coercive field. Comparing with previous single-component organic ferroelectrics, such as TCAA (spontaneous polarization $P_s = 0.2 \mu\text{C cm}^{-2}$),^{7,8} thiourea ($P_s = 3.2 \mu\text{C cm}^{-2}$),⁹ and TEMPO ($P_s = 0.5 \mu\text{C cm}^{-2}$),¹⁰ croconic acid demonstrates excellent ferroelectricity at room temperature, and its spontaneous polarization P_s is up to $21 \mu\text{C cm}^{-2}$. Its spontaneous polarization changes a little when the temperature changes between 5 and 400 K, and its dielectric constant has a slight increase when heated up to 380 K.⁶ Some papers have already studied the polarization of croconic acid in theory and experiments,^{11–17} however, its microscopic ferroelectric mechanism in terms of the electronic structure and the molecular structure are worth further studying. Up to now, we have not found references studying the thermal properties of the ferroelectricity of croconic acid in theory.

This paper is organized in the following way. In Sec. II, we give the details of our calculations. In Sec. III A, we investigate the microscopic ferroelectric mechanism of croconic acid by the electronic structure, the deformation charge density and Bader charge. The simulation of the protons transfer and the discussion on the polarization reversal are presented in Sec. III B. In Sec. III C, we calculate the polarization by Berry phase, and research the ferroelectric phase transition of croconic acid by Monte Carlo method. Finally, a summary is given in Sec. IV.

II. THEORETICAL METHODS

The molecule of croconic acid possesses a pentagon of five carbon atoms, with two hydroxyl groups orienting in

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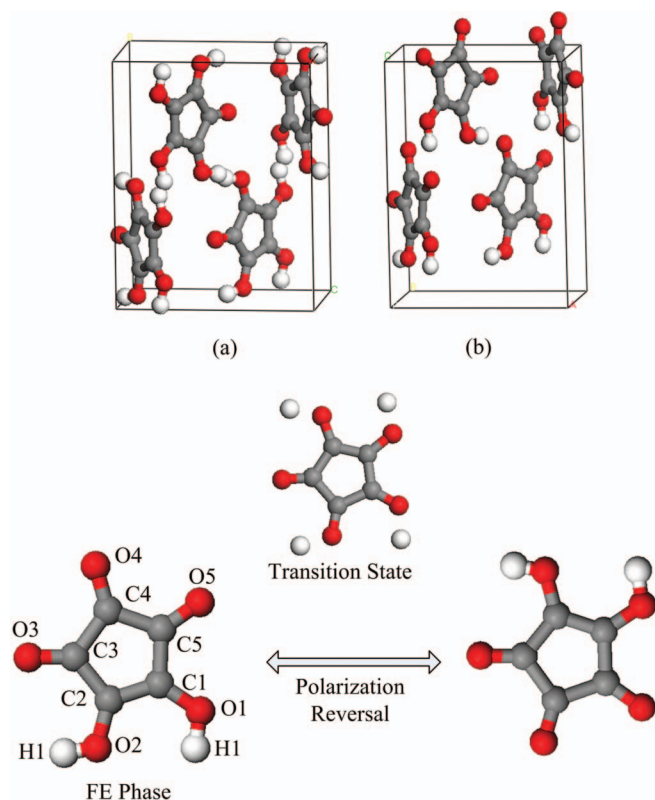


FIG. 1. Structural comparisons of orthorhombic croconic acid crystal with different space group number. (a) The orthorhombic system with space group *Pbcm* with the inversion center, with protons in two-fold positional disorder, which is fabricated to be used to construct the paraelectric (PE) phase. (b) The orthorhombic system with space group *Pca21* without the inversion center.

the same direction attached to two adjacent carbon atoms, which is plotted in Fig. 1. Croconic acid crystal belongs to orthorhombic phase with space group *Pca21* without the inversion center. In the present work, we calculated the electronic structure, charge density, and deformation charge density of croconic acid by using Vienna *ab initio* Simulation Package (VASP). The first principle calculations were performed using density functional theory (DFT) with generalized gradient approximation (GGA). In self-consistent calculations, the Γ -centered Monkhorst-Pack scheme was used to generate the k points in the Brillouin zone.¹⁸ A plane-wave cut-off of 500 eV and a k -point grid of $6 \times 4 \times 3$ were used. The convergence criterion of the total energy is 10^{-4} eV and the force convergence criterion is taken to be 0.01 eV/Å for the optimization of the lattice parameters. Based on the experimental data (collected from X-ray diffraction in the Cambridge Crystallographic Data Centre) of $a = 8.711$ Å, $b = 5.169$ Å, and $c = 10.962$ Å, the optimized lattice parameters are $a = 9.035$ Å, $b = 5.250$ Å, and $c = 11.043$ Å. The deviations of optimized lattice a , b , and c from the experimental values are 4%, 2%, and 1%, respectively. By adding van der Waals type functionals, the optimized lattice parameters can be improved, as demonstrated by Di Sante *et al.*¹¹ In the present work, the calculations were performed at the optimized lattice parameters by GGA, since we found that the calculated polarization by the GGA is in good agreement with the experimental value.

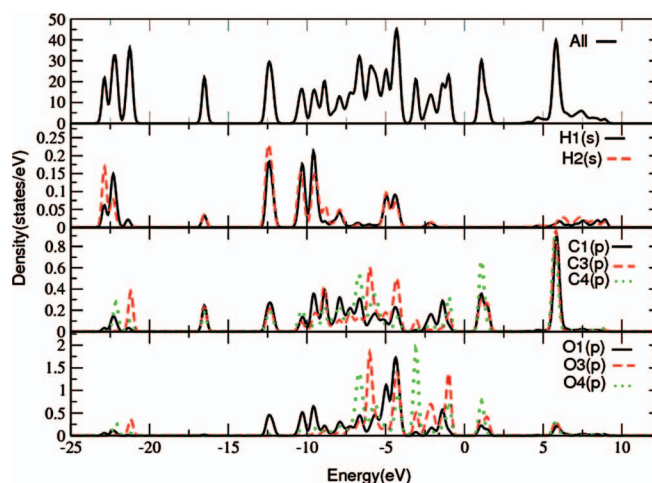


FIG. 2. The DOS and PDOS of croconic acid.

III. RESULTS AND DISCUSSION

A. Ferroelectric mechanism

The total density of states (DOS) and the partial DOS (PDOS) of croconic acid are shown in Fig. 2. The characteristic peaks of the DOS represent that the energy bands are all flat and not seriously dispersive, which shows the typical of the electronic structure of a molecular crystal.¹⁹ The calculated indirect band gap is about 1.82 eV. The valence bands mainly are consisted of 1s orbital of H, 2p orbitals of O and C, and the bottom of the conduction bands only contain 2p orbitals of O and C. As shown in Fig. 2, the C atoms and its adjoining O atoms exhibit the same PDOS peaks in the valence band and the bottom of the conduction band, which means that the 2p orbitals of these C and O atoms mix heavily and form the strong covalent bonds between them. The PDOS of the H atoms, which is nearly zero at the bottom of the conduction band, shows the ionic bond characteristic between the H atoms and their adjoining O atoms.

The electron deformation density of the molecule is the difference between the electron density of the molecule and that of the independent atom model (IAM) $\rho - \rho_{IAM}$.²⁰ The electron deformation density illustrates the nature and the extent of polarization ensuing in the molecule. The deformation density is projected to the plane determined by the pentagon of the five C atoms, as shown in Fig. 3(b), and its electron deformation density range is from -0.6 to 0.8 eÅ⁻³. The electron deformation map shows that the croconic acid molecule is connected with four nearest molecules by four intermolecular hydrogen bonds O—H...O, respectively. In Fig. 3(b), we find that the valence electrons of O atoms of the keto group CO are obviously polarized towards the nearest H atoms of the enol group OH by the influence of the intermolecular hydrogen bond, and the partly exposed hydrogen nuclear draws obviously the valence electrons of the O atom of the nearest neighbor molecule. The electron deformation density and the electron density maps [show in Fig. 3(a)] suggest there are five covalent groups CO and a π conjugation bond consisted of the five CO groups. The electron-withdrawing groups “C4 = O4, C5 = O5” with high

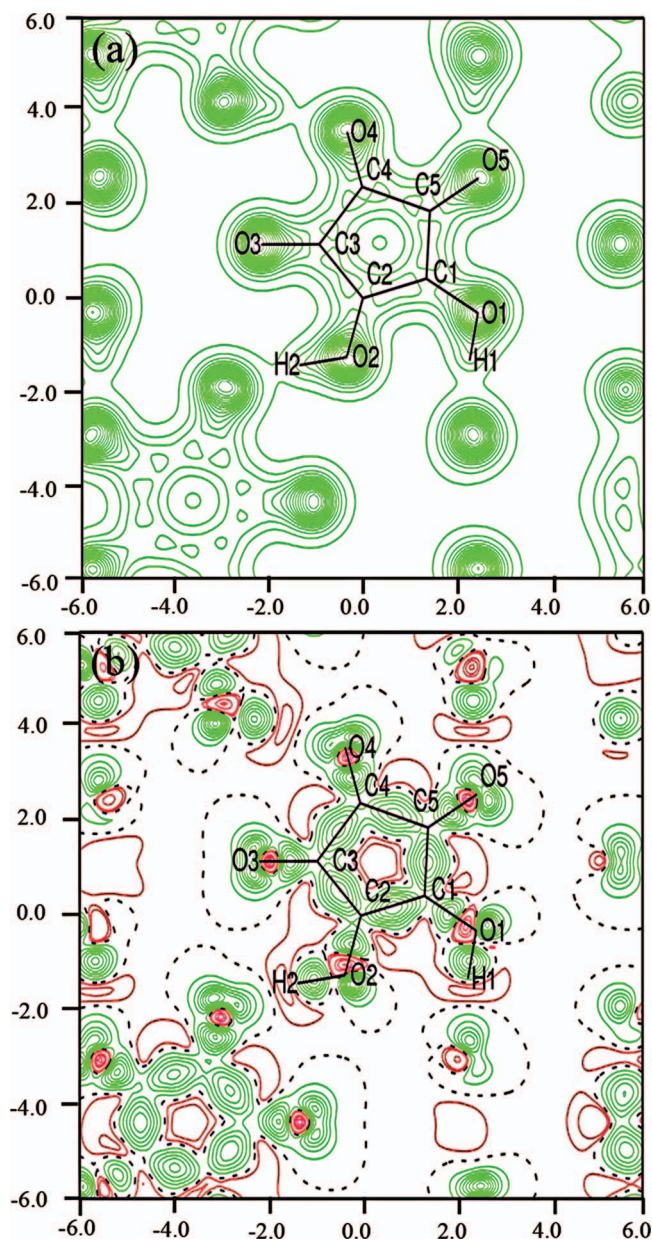


FIG. 3. (a) The electron density map of croconic acid with contour levels of $0.9 \text{ e}\text{\AA}^{-3}$. (b) Deformation density maps of croconic acid with contour levels of $0.1 \text{ e}\text{\AA}^{-3}$. Green lines are positive, red lines negative, black lines are zero.

electron affinity gain electron from the electron-releasing groups “O1–H1, O2–H2”. In order to see clearly the charge transfer (CT), we have calculated the net Bader charge²¹ on each atom, as shown in Table I. From Table I, we find that the charge of the electron-releasing groups H1O1C1 (H2O2C2) and the electron-withdrawing group C4O4 (C5O5) are about +0.2 electrons and –0.2 electrons, respectively, it indicates there are about 0.2 electrons transferring from the electron-releasing group to the electron-withdrawing group.

The CT from the electron-releasing group to the electron-withdrawing group appears, and the π conjugation pentagon acts as a good medium through which the charge can be transferred. Thus the intermolecular CT through hydrogen bonds will favor a head-to-tail (non-centrosymmetrical) arrangement of molecular electric dipoles, and thereby enhance the polarization. This is one of the reasons why the crystalline polarization is higher than the sum of molecular dipoles calculated by Horiuchi *et al.*⁶

B. Polarization reversal

The polarization reversal is a very important property for ferroelectrics, and it is hard to realize because of severe steric hindrance due to the molecular anisotropy. We try to simulate the proton transfer to study the polarization reversal. In the simulation, we set the C and O atoms fixed, the H atoms were relaxed under the external electric field until forces on the atoms were lower than $0.01 \text{ eV/\AA}$. When the applied electric field is increased to $5.0 \times 10^{-3} \text{ V/\AA}$ along c axis, the crystal structure can be topologically reversed through proton transfer. The changes of the hydrogen bonds are O1–H1···O5 → O1···H1–O5, O2–H2···O4 → O2···H2–O4, and the bond lengths are almost exchanged. The simulated saturated electric field $5.0 \times 10^{-3} \text{ V/\AA}$ is much bigger than the experimental value $4 \times 10^{-4} \text{ V/\AA}$.⁶ Frankly, we do not know the reason for such big error. In the following work, we will try to study the problem by relaxing the H and O atoms and just fixing the C atoms.

The process of the polarization reversal is accompanied with the tautomerism happening between the keto and the enol forms: $\text{C}=\text{O} \cdots \text{H}-\text{O}-\text{C} \leftrightarrow \text{C}-\text{O}-\text{H} \cdots \text{O}=\text{C}$. During such a tautomerism, the molecule of croconic acid also plays a dual role of proton donor and proton acceptor. The H atoms transfer closely to the O atoms of nearest molecules. At the same time, the electron density of π -bond system is redistributed according to the proton transfer. Croconic acid is a kind of β -diketone enolates and its hydrogen bond donor and acceptor are connected through π bonds. This hydrogen bond is so-called resonance-assisted hydrogen band (RAHB).^{22,23} This RAHB strengthens the intramolecular and intermolecular interactions, as well as outstanding ferroelectricity of croconic acid. However, not all hydrogen bonds can be changed under electric field and benefit the ferroelectricity of materials. Hydrogen bonds are classified as strong, moderate, and weak by Jeffrey.²⁴ For the hydrogen bond containing H1, the calculated bond length H···O, H–O, the distance between these two O atoms and the bond angle are $1.619 \text{ \AA}$, $1.032 \text{ \AA}$, $2.636 \text{ \AA}$, and 167.551° , respectively. For the hydrogen bond containing H2, the calculated values of the same physical quantities as the above are $1.671 \text{ \AA}$, $1.022 \text{ \AA}$, $2.685 \text{ \AA}$, and 171.244° , respectively. According

TABLE I. Net Bader charge on various atoms in croconic acid molecules.

	H1	H2	O1	O2	O3	O4	O5	C1	C2	C3	C4	C5
Net Bader charge	0.675	0.689	–1.139	–1.144	–1.054	–1.080	–1.074	0.724	0.646	0.965	0.9132	0.869

to the criterion presented by Jeffrey,²⁴ the hydrogen bonds of croconic acid belong to the strong moderate hydrogen bonds. It is well known that the strong hydrogen bond restricts the movement of the proton (H atom) of the hydrogen bond, so the polarization reversal cannot be realized in the materials with strong hydrogen bond. The weak hydrogen bond can change easily under applied electric field; however, it cannot ensure big polarization. The moderate hydrogen bond is mainly the electrostatic interaction, this kind of hydrogen bond allow the movement of the proton along the hydrogen bond to realize polarization reversal, also can introduce a high polarization. Recently, some organic ferroelectrics²⁵ of chains via moderate hydrogen bonds, such as 2-phenylmalondialdehyde (PhMDA)²⁶ and cyclobutene-1,2-dicarboxylic acid (CBDC),²⁷ are discovered.^{24–26} The polarization and the polarization reversal of these organic ferroelectrics are also from the contribution of the hydrogen bonds.¹² Moderate hydrogen bond is a very important clue to find new organic ferroelectrics.

C. Polarization and ferroelectric phase transition

The molecular dipole moment is an important factor for polarization of material. The molecule of croconic acid with two hydroxyl groups orienting in the same direction has a large dipole moment of 9–10 D (1 D = 3.336×10^{-30} C m).²⁸ In gas phase, the other isomer of croconic acid with two hydroxyl groups orienting in opposite direction, has a smaller dipole moment.²⁹ Comparing with this isomer, the molecule of croconic acid with hydroxyl groups orienting in the same direction has a larger dimerization binding energy via hydrogen bonds, which makes itself has some advantages during the crystallization. Some organic ferroelectrics, such as TCAA,³⁰ also have considerable molecular dipole, however, unlike croconic acid, their opposing orientations of molecules in crystal make most of the dipole moments counterbalanced. That is one of reasons why organic ferroelectrics are rare.

In a finite system, it is appropriate that polarization is defined as the total dipole moment divided by the volume of the system. However, in an infinite periodic solid, different electric polarization vector can be acquired if different unit cell is chosen. The modern theory of polarization^{31,32} points out that the change of the polarization can be interpreted as a displacement of the center of charge of the Wannier functions, and the problem of calculating polarization in periodic crystals can be solved by using the Berry phase of the electronic wavefunction. In this paper, we employ Berry phase method^{31,33} to calculate spontaneous polarization of croconic acid. In order to calculate the polarization and study the contributions of atoms to the polarization, a centric high-symmetry (or paraelectric) structure will be adopted as zero polarization reference. We know that every proton has two possible positions corresponding to positive polarization and reverse polarization, respectively. We place the protons in the positions in the two-fold positional disorder, and get the positions of the other atoms with CASTEP by setting the symmetric group *Pbcm*, as shown in Fig. 1(a). Aside from the remarkable proton transfer distortion, there are small distortion of oxygen atomic dis-

placements nearby the protons, and negligible displacements of carbon atoms and other oxygen atoms. Using partial mode analysis of the contributions to the ferroelectricity,¹² we calculate the contribution of protons and oxygen atoms nearby the protons to the ferroelectricity. First, we only consider the effect of the protons transfer on polarization with keeping the other atoms in their centrosymmetric position. The calculated spontaneous polarization is $P_s = 22.3 \mu\text{C cm}^{-2}$ along the *c* axis. Then, the displacements of oxygen atoms nearby the protons (i.e., O1, O2, O4, and O5) from the centric structure to the polar structure are also taken into consideration, and the polarization introduced by the oxygen atoms and protons is calculated. The calculated spontaneous polarization is increased to $24.7 \mu\text{C cm}^{-2}$ along the *c* axis, which is in good agreement with the experimental value about $21 \mu\text{C cm}^{-2}$. About the more detailed study on the contributions of the displacements of the atoms to the polarization, readers can see Ref. 11.

Now, we focus to study the thermal properties of the ferroelectricity of croconic acid. In Sec. III A, we have explained that the arrangement of the molecular electric dipoles one by one in chains, and forms a superdipole along the *c* axis. Only considering the nearest neighbor interaction of the electric dipole moments of the molecules, the Hamiltonian of ferroelectric can be written as

$$H = - \sum_{i,j} K P_i P_j - \sum_i E P_i, \quad (1)$$

where P_i and P_j are the electric dipole moment of the *i*th and *j*th dipole, E is an external electric field and K denotes the coupling coefficient between the two nearest dipoles. Our objective is to simulate phase transition temperature, so we let the external electric field to be zero. Defining a line connecting the two possible positions of proton as an *x* axis, we choose the middle point of these two positions as the original point *O* of the *x* axis. Thus the coordinates of the two possible positions are set to be $x = -1$ and $x = 1$, respectively. $x = 0$, $x = 1$, and $x = -1$ are correspond to the centric, positive, and negative polar configuration, respectively. Then we calculate the polarization of some structures that fall in between the centric (paraelectric) structure ($x = 0$) and the polar (ferroelectric) distorted structure ($x = \pm 1$). In Fig. 4, we show the variation of the energy difference ΔE and of polarization as a function of the amplitude of the polar distortion. From Fig. 4, we find the bi-stable energy profile characteristic of the ferroelectric material with an energy barrier $\Delta E = 1.48$ eV between centric structure and polar structure in one unit cell.

From Fig. 4, we find that the electric dipole moment (P) is approximately proportional to relative displacement of the proton (x). So, the Hamiltonian can be simplified as

$$H = - \sum_{i,j} k x_i x_j. \quad (2)$$

In each unit cell of croconic acid, there are 4 parallel dipole-chains, and each chain has two dipoles. The coefficient can be gotten by $k = \Delta E/8 = 1.48 \text{ eV}/8 = 0.185 \text{ eV}$. In the present simulations, we use periodic boundary condition to erase the influence of the boundary effect upon the simulated calculations. The relative displacement of the proton (x) takes

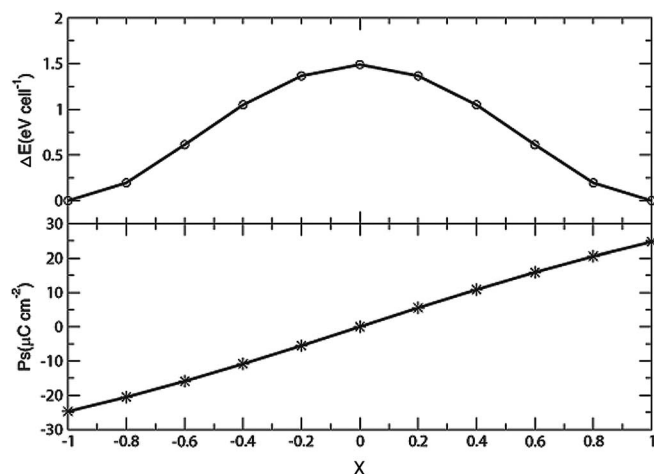


FIG. 4. Variation of total energy ΔE (top) and of spontaneous polarization P_s (bottom) as a function of relative displacement of the proton (x) between centric ($x = 0$) and polar ($x = \pm 1$) configurations.

new value with the aid of Metropolis importance sampling,³⁴ and the accepting probability of the new position can be written as

$$p = \begin{cases} 1 & \Delta E < 0 \\ \exp(-\Delta E/k_B T) & \Delta E \geq 0 \end{cases}, \quad (3)$$

where ΔE denotes the energy difference between the new and old states of the system, T is the temperature and k_B is the Boltzmann constant.

We adopt the Metropolis method to sample. This method ensures that the microscopic state that contributes the most to the statistic mean value will not be omitted. The equilibrium state of the system is achieved by using a Markov process. The present calculations show that the system with 2000 unit cells has achieved an equilibrium state after much less than 200 000 Monte Carlo steps, so we abandon the first 200 000 Monte Carlo steps, then calculate the statistic mean value of the last 100 000 MC steps by using the relations as follows:

$$\chi = \frac{\langle p^2 \rangle - \langle p \rangle^2}{k_B T}, \quad (4)$$

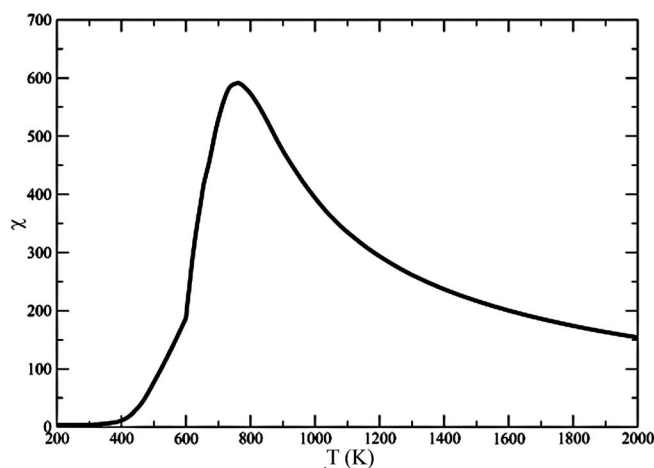


FIG. 5. The polarizability curves versus the temperature.

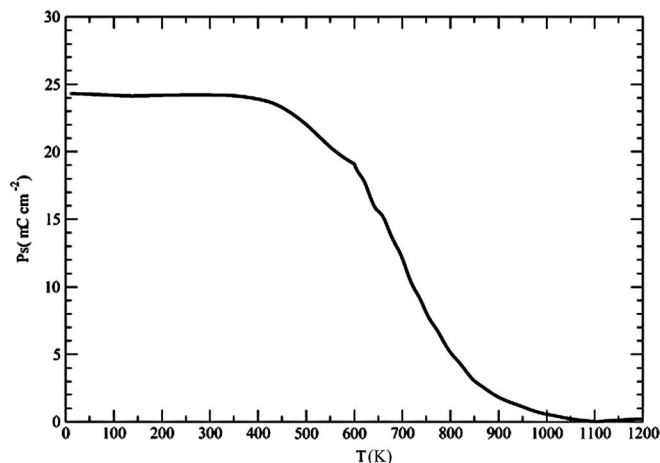


FIG. 6. The spontaneous polarization curves versus the temperature.

where χ is the polarizability of the system and P is the polarization.

The temperature dependence of polarizability by Monte Carlo simulations is illustrated in Fig. 5. The polarizability has a peak at $T = 756$ K corresponding to the Curie temperature (T_c). The strong coupling between dipoles results in the high T_c . We plot the spontaneous polarization against temperature in Fig. 6. Below 400 K, the change of polarization ΔP is less than $1 \mu\text{C cm}^{-2}$, which is consistent with the experimental results⁶ ($\Delta P = 1\text{--}1.5 \mu\text{C cm}^{-2}$) very well. Croconic acid decomposes at about 450 K,⁶ so it is regrettable that we cannot judge the simulated T_c (756 K) by the experimental result.

IV. CONCLUSION

We have performed the first-principles calculations of the ferroelectric properties of a single-molecular organic material croconic acid within DFT in the GGA. The spontaneous polarization of croconic acid has been obtained from the wave functions by Berry phase method, and the calculated value is in good agreement with the experiments. Based on the analysis of the electronic structure, the crystal configuration, molecular structure, it is found that the origin of the large polarization is the CT due to the strong “push-pull” effect of electron-releasing and -withdrawing groups along the hydrogen bond. The proton in the moderate hydrogen bond can be transferred when external electric field applied, and the polarization reversal is realized through the transfer of the proton. By using one-dimensional ferroelectric Hamiltonian model and Monte Carlo method, the simulated polarizability and the temperature dependence of the spontaneous polarization are in good agreement with the experimental data. This indicates indirectly that the one-dimensional ferroelectric model and the nearest neighbor interaction approximation of the dipoles of the molecules are suitable for croconic acid.

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